Prabhavit Gupta

Python Developer: LangGraph Orchestration, Agent Workflows & Tracing

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Summary

Python Developer with 3 years of commercial software engineering experience specializing in multi-step AI workflow orchestration and agent systems. Deep practical expertise architecting reliable graph-based execution loops using LangGraph and LangChain, implementing stateful checkpointing across PostgreSQL, Redis, and SQLite backends. Proven ability to design resilient human-in-the-loop pause-and-resume routines, automated timeout and fallback patterns, and end-to-end distributed tracing via OpenTelemetry. Dedicated to building observable, production-grade agent platforms that execute long-running asynchronous workflows deterministically.

Experience

FlowSpace.co • *Lead Developer & AI Systems Architect (Freelance)*
Mar 2024 - Aug 2026 • London, UK (remote)

- Engineered multi-step Python agent workflows and orchestration graphs coordinating document extraction, entity validation, and database synchronization for 4,500+ active enterprise users.
- Implemented stateful checkpointing with PostgreSQL and Redis backends, ensuring workflow state persistence across asynchronous task boundaries and multi-turn user interventions.
- Designed human-in-the-loop interrupt and approval gates within workflow execution graphs, allowing domain experts to inspect and adjust intermediate extraction payloads before final commit.
- Architected resilient execution strategies featuring automated exponential backoff retries, node-level timeouts, and graceful model fallback chains across Gemini, Claude, and OpenAI APIs.
- Integrated OpenTelemetry instrumentation and structured logging across orchestration steps, tracking trace spans, invocation latencies, and token expenditures in production.
- Maintained automated regression test suites and CI/CD pipelines via GitHub Actions and Docker containerization.
- *Tools: Python, LangGraph, LangChain, asyncio, PostgreSQL, Redis, SQLite, OpenTelemetry, Docker, Linux*

Urlink • *Project Manager & Solutions Developer (Freelance)*
Apr 2025 - Jul 2025 • Barcelona, Spain (remote)

- Developed Python automation scripts and asynchronous event handlers connecting external enterprise APIs with PostgreSQL data pipelines.
- Designed state transition logic and schema validation gates, preventing corrupted or incomplete payloads from propagating through multi-stage data jobs.
- Produced comprehensive architecture documentation, API schemas, and sequence diagrams for asynchronous integration services.
- *Tools: Python, asyncio, REST APIs, PostgreSQL, Supabase, TypeScript*

Independent Technical Consultant • *AI Automation & Integration Developer*
Sep 2023 - Mar 2024 • Venice, Italy

- Built custom automated data extraction agents and webhook dispatchers in Python, handling scheduled batch processing and real-time event triggers.
- Implemented circuit-breaker and retry logic for high-frequency external API calls, reducing integration dropouts and ensuring idempotent transaction handling.
- Authored technical implementation specifications, test scripts, and system health monitoring procedures for client engineering teams.
- *Tools: Python, Webhooks, REST APIs, Supabase, Linux*

Tractionboard.co • *Low-Code Developer (Internship)*
Aug 2023 - Oct 2023 • Utrecht, Netherlands (remote)

- Integrated LLM API calls and validation routines into core customer workflows, creating automated test scenarios for user onboarding funnels.
- *Tools: Python, OpenAI API, Sendgrid, REST APIs*

Education

Ca' Foscari University of Venice • *Bachelor's in Philosophy, International Studies & Economics, Minor in Entrepreneurship*
Oct 2020 - Jun 2024 • Venice, Italy

Certifications

Microsoft: Introduction to building AI agents • Microsoft Copilot Studio Agent Academy • Microsoft Power Automate • Tray.AI Build Practitioner (Sep 2024) • Tray.AI Build Foundations

Additional Information

- **Work authorization:** European Long Term Residence Permit (Italy), unrestricted EU work authorization (Partita IVA / B2B ready). No sponsorship required.
- **Languages:** English (fluent / C1 professional), Hindi (native), Italian (beginner).
- **Availability:** Available immediately for full-time remote roles across European time zones.

Core Skills

- **AI Orchestration & Workflow Engines:** LangGraph (nodes, edges, state graphs), LangChain, LangSmith, human-in-the-loop workflows, interrupt/resume mechanics.
- **Python & Async Development:** Python 3, asyncio, concurrent tasks, worker queues, custom decorators, error handling and resilient fallbacks.
- **State Persistence & Checkpointing:** PostgreSQL checkpointer, Redis checkpointer, SQLite checkpointer, state serialization, schema versioning, transaction boundaries.
- **Observability & Reliability:** OpenTelemetry distributed tracing, span instrumentation, structured logging, latency profiling, token tracking, retry mechanisms with exponential backoff.
- **Cloud & Engineering Practices:** Docker containerization, Linux environments, Git, GitHub Actions CI/CD, automated integration testing, RESTful API design.